

Pre-design solution: Тестовый отказ системы

Region: Новосибирская область **Greenhouse type:** year-round **Crop:** tomato **Target yield:** 2000.0 t/year **Generated:** 2026-06-17 05:46

1. Site inputs and analysis

Target yield per plot area: 4000.0 kg/m²/year $\triangle$ Target yield is very high — intensive technology required.

Climate parameters (per Russian SP 131.13330):

Parameter	Value
Design winter temperature (t5)	-37.0 °C
Design summer temperature	25.0 °C
Heating degree-days	6076.0
Heating period	227 days
Snow load	2.4 kPa
Wind load	0.3 kPa
Solar radiation (winter)	1.5 MJ/m ² /day

2. Design solution (variant failed_v1)

Rationale: This design proposes two compact year-round glass greenhouse blocks (Block A and Block B) on a very constrained 25 × 20 m plot in the Novosibirsk region, targeting tomato trellis culture. Glass cladding is chosen to maximise light transmittance ($\tau \approx 0.88$), which is critical given the region's low winter solar radiation. Each block is a single-span (angar) layout because the plot width cannot accommodate multi-span block configurations while maintaining the mandatory 6 m inter-block clearance for year-round greenhouses per SP 107.13330 clause 4.4. Ridge height is set to 5.5 m as required for trellis tomato culture. It must be noted that the plot (500 m²) is physically unable to deliver 2 000 t/year of tomatoes under any greenhouse configuration — the agronomic carrying capacity of this site is many orders of magnitude below the stated target yield; the project brief contains a fundamental contradiction between plot size and yield target that cannot be resolved by layout changes alone.

Total growing area: 72 m² **Complex footprint (with auxiliary):** 500.0 m²

Greenhouse blocks

- **Block A** — 6.0 × 6.0 m (area 36 m²)
- Heights: eave 2.4 m, ridge 5.5 m
- Layout: angar (single-span), 1 spans of 6.0 m
- Roof: straight, slope 45.0%
- Covering: glass ($\tau=0.88$), glass 4.0 mm - Opaque structures share: 15.0%
- **Block B** — 6.0 × 6.0 m (area 36 m²)
- Heights: eave 2.4 m, ridge 5.5 m
- Layout: angar (single-span), 1 spans of 6.0 m
- Roof: straight, slope 45.0%
- Covering: glass ($\tau=0.88$), glass 4.0 mm - Opaque structures share: 15.0%

Complex structural parameters

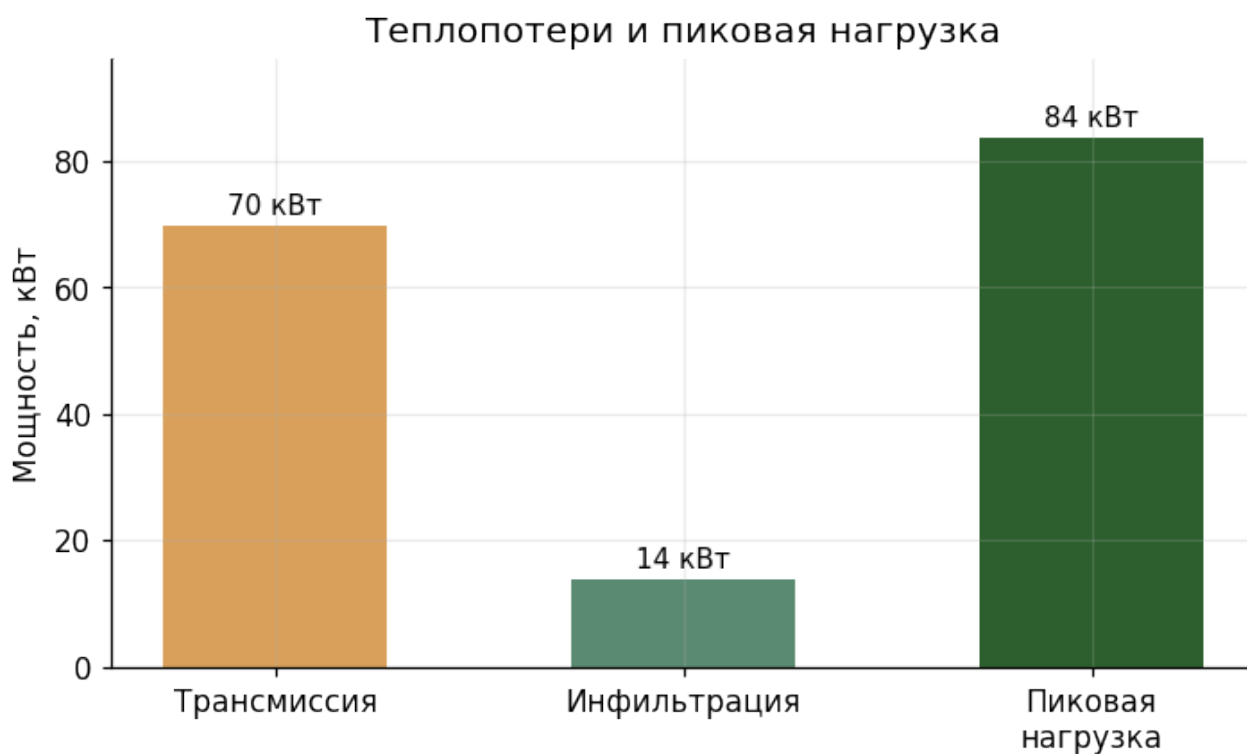
Parameter	Value	SP Clause
Plinth height	0.4 m	5.8
Foundation rise above soil	0.4 m	5.9
Territory fence height	1.8 m	4.16

Auxiliary zones

- **Boiler house** — 12.0 m² (Heating plant and utility room)
 - **Tool and fertiliser storage** — 8.0 m² (Storage for tools, agrochemicals and fertilisers)
 - **Packaging and sorting area** — 8.0 m² (Post-harvest packaging and product sorting)
 - **Staff facilities** — 7.0 m² (Changing rooms, sanitary facilities and rest room for personnel)
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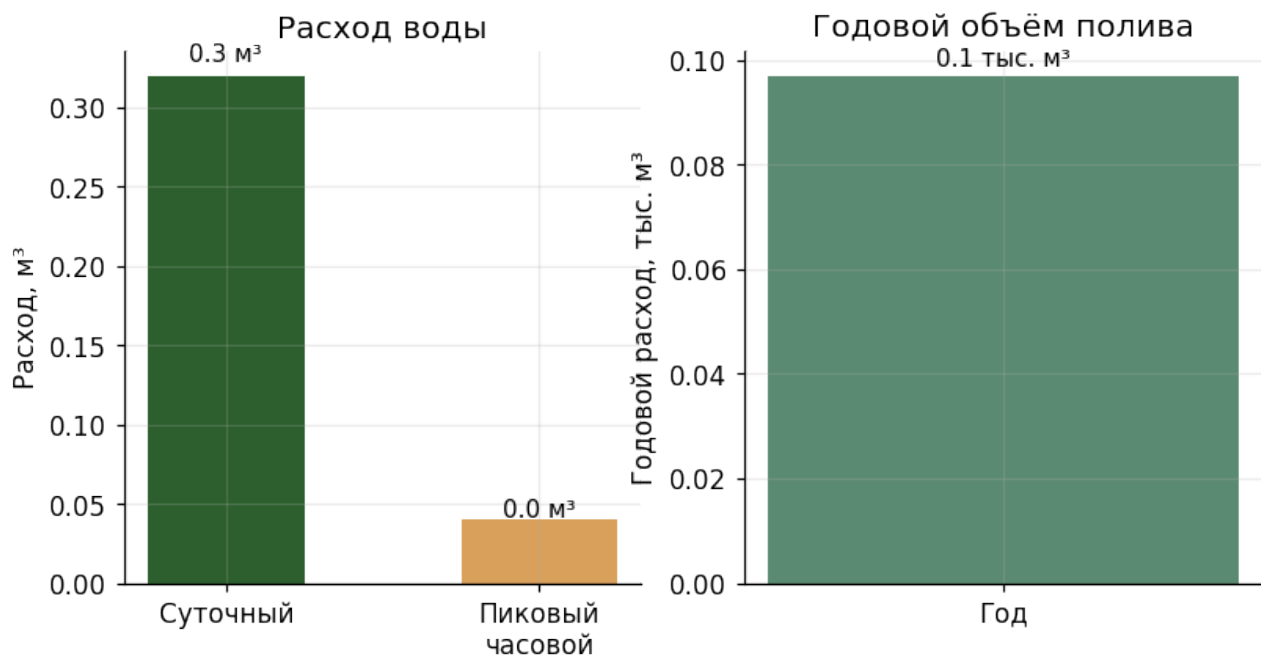
3. Engineering calculations

3.1. Heat supply



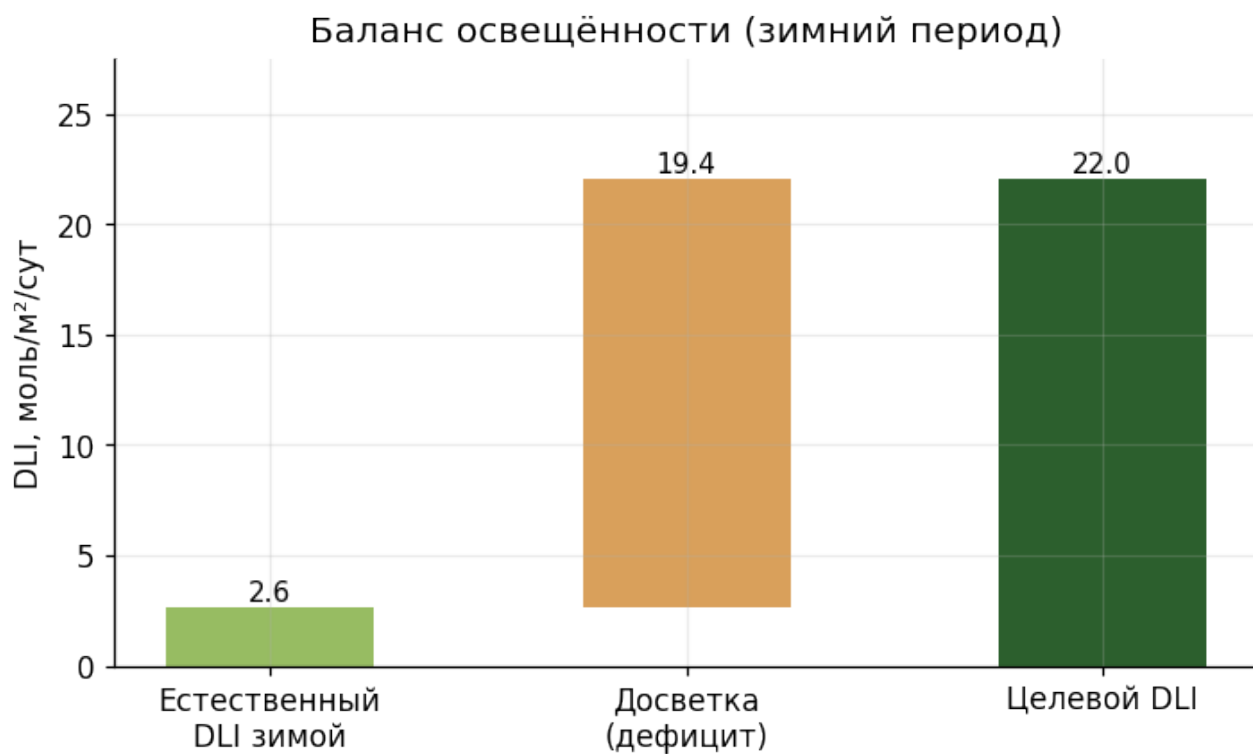
- Design temperature difference: **55.0 °C**
- Envelope area: **198.0 m²**
- Overall heat transfer coefficient U: **6.4 W/(m²·K)**
- Transmission losses: **69.7 kW**
- Infiltration losses: **13.9 kW**
- **Peak heating load: 83.6 kW**
- Annual heat demand: **221.7 MWh**
- Coolant temperature: **95.0 °C** (SP 7.9 ≤150)
- Lower-zone heat share: **45.0%** (SP 7.13 ≥40)

3.2. Water supply



- Daily demand: **0.32 m³/day**
- Peak hourly: **0.04 m³/h**
- Annual: **97 m³/year**
- Irrigation method: Drip (default)
- Reliability category: **II** (SP 6.14)
- Hose service radius: **40.0 m** (SP 6.8 ≤45)

3.3. Lighting and supplemental



- Target DLI: **22.0 mol/m²/day**
- Natural winter DLI: **2.6 mol/m²/day**
- **Supplemental lighting required:** installed 266.2 W/m², consumption 69612 kWh/year
- Aisle floor illuminance: 8.0 lx (SP 8.3 ≤10)

3.4. Annual energy consumption

Годовое энергопотребление: 291 МВт·ч



3.5. Ventilation

- Summer target air changes per hour: **60.0 h⁻¹**
- Vent opening share of envelope: **15.0%** (SP 7.18 ≥ 20 , ≥ 10 north of 60°N)
- Vent opening share of floor: 41.2%
- Forced ventilation: **not required**

3.6. Structural loads

- Snow load on roof: **278.0 kN** ($\gamma=1.4$)
- Wind load on walls: **28.0 kN** ($q_{10}=1.0$, $q_2=0.6$)
- Trellis load: 150.0 N/m² ($\gamma=1.3$)

Overload factors and normative values per SP 107.13330 clause 5.14.

3.7. Geotechnical conditions and foundation

- Foundation soil: **clay**
- Groundwater depth: **0.5 m**
- Recommended foundation type: **pile**
- Minimum embedment depth: **3.0 m**

- Perimeter drainage: **required**
- Waterproofing grade: **enhanced**

Clay with high groundwater table — frost-heaving soil. Pile foundation embedded below the freezing depth; enhanced waterproofing and mandatory perimeter drainage required.

Preliminary recommendations. A proper engineering-geological survey per SP 47.13330 is required for working documentation.

4. SP 107.13330 compliance check

Rules checked: **35**

ERROR — ENG.5-yield-feasibility

Площадь выращивания × норма культуры должна покрывать минимум 90% целевой урожайности из ТЗ. Иначе участок физически не вмещает заявленный объём продукции.

- Actual: вмещает ~4 т/год (tomato @ 50.0 кг/м² × 72 м²)
- Required: >= 1800 т/год (90% от ТЗ 2000.0 т)
- **Source** (SP 107.13330, original Russian text): Инженерная проверка (не из СП): достижимость целевой урожайности

Generated by agro-greenhouse-designer. Outputs are pre-design estimates and require validation by a qualified engineer before detailed documentation.